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Ethernet and Fibre Cable Types Cheat Sheet

Ethernet Cables (Copper)

Cat5e (Category 5 Enhanced)

Speed: Up to 1 Gbps **Distance**: 100 meters (328 feet) **Frequency**: 100 MHz

Best for:

- Basic network connections
- Home and small office networks
- Budget-conscious installations

Pros: Inexpensive, widely available **Cons**: Limited speed, older standard

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Cat6 (Category 6)

Speed: Up to 1 Gbps (10 Gbps up to 55 meters) **Distance**: 100 meters (328 feet) **Frequency**: 250 MHz

Best for:

- Modern business networks
- Future-proofing installations
- Areas with electrical interference

Pros: Better performance than Cat5e, reduced crosstalk, supports short 10G runs **Cons**: More expensive than Cat5e, thicker cable

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Cat6a (Category 6 Augmented)

Speed: Up to 10 Gbps **Distance**: 100 meters (328 feet) **Frequency**: 500 MHz

Best for:

- High-performance networks
- Data centers
- Server connections
- Full 10 Gigabit runs

Pros: Full 10G support at 100m, excellent shielding **Cons**: Thicker and stiffer, more expensive

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Cat7 (Category 7)

Speed: Up to 10 Gbps **Distance**: 100 meters (328 feet) **Frequency**: 600 MHz

Best for:

- High-interference environments
- Professional installations
- Shielded applications

Pros: Excellent shielding, high frequency support **Cons**: Requires special connectors, overkill for most applications

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Cat8 (Category 8)

Speed: Up to 40 Gbps **Distance**: 30 meters (98 feet) **Frequency**: 2000 MHz

Best for:

- Data center connections
- High-speed server links
- Short-run high-performance applications

Pros: Extremely high speeds, future-proof **Cons**: Very expensive, limited distance, specialized use only

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Fibre Optic Cables

Single-Mode Fibre (SMF)

Core size: 8.3-10 microns **Distance**: Up to 40+ km **Speed**: Up to 100+ Gbps

How it works: Uses laser light through a very thin core

Best for:

- Long-distance connections
- Campus networks
- ISP backbone connections
- High-speed WAN links

Pros: Longest distances, highest speeds, low signal loss **Cons**: More expensive equipment, requires precise alignment

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Multi-Mode Fibre (MMF)

OM1 (Orange jacket)

Core size: 62.5 microns **Distance**: Up to 550m (1 Gbps), 300m (10 Gbps) **Speed**: Up to 10 Gbps

Best for: Legacy installations, shorter runs

OM2 (Orange jacket)

Core size: 50 microns **Distance**: Up to 550m (1 Gbps), 550m (10 Gbps) **Speed**: Up to 10 Gbps

Best for: Standard multi-mode applications

OM3 (Aqua jacket)

Core size: 50 microns **Distance**: Up to 1000m (1 Gbps), 300m (10 Gbps) **Speed**: Up to 40 Gbps

Best for: Modern LAN applications, data centers

OM4 (Aqua jacket)

Core size: 50 microns **Distance**: Up to 1100m (1 Gbps), 400m (10 Gbps) **Speed**: Up to 100 Gbps

Best for: High-performance data centers, future-proofing

OM5 (Lime green jacket)

Core size: 50 microns **Distance**: Similar to OM4 **Speed**: Up to 100+ Gbps

Best for: Wavelength division multiplexing, advanced applications

Multi-Mode Pros: Less expensive than single-mode, easier to terminate **Multi-Mode Cons**: Shorter distances than single-mode, lower maximum speeds

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Quick Comparison - Ethernet

Cable Type	Max Speed	Distance	Frequency	Best Use
Cat5e	1 Gbps	100m	100 MHz	Basic networks
Cat6	1-10 Gbps	100m	250 MHz	Modern business
Cat6a	10 Gbps	100m	500 MHz	High performance
Cat7	10 Gbps	100m	600 MHz	Shielded environments
Cat8	40 Gbps	30m	2000 MHz	Data centers

Quick Comparison - Fibre

Fibre Type	Core Size	Max Distance	Max Speed	Cost
Single-Mode	9 microns	40+ km	100+ Gbps	High
OM1	62.5 microns	550m	10 Gbps	Low
OM2	50 microns	550m	10 Gbps	Low
OM3	50 microns	1000m	40 Gbps	Medium
OM4	50 microns	1100m	100 Gbps	Medium
OM5	50 microns	1100m	100+ Gbps	Medium-High

Choosing the Right Cable

For Ethernet:

- **Cat5e**: Budget installs, basic needs
- **Cat6**: Standard business choice
- **Cat6a**: Future-proofing, 10G requirements
- **Cat8**: Data center short runs only

For Fibre:

- **Single-Mode**: Long distances, highest performance
- **OM3/OM4**: Modern data centers and LANs
- **OM5**: Latest standard for advanced applications

Consider: Distance requirements, speed needs, budget, and future expansion plans